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Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Time: 03 Hours

Code: 101503 ***Subject: Geotechnical Engineering - I*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[14]**

(a) Specific gravity refers to the ratio of the weight of soil solids to
- (i) Weight of equal volume of water
- (ii) Unit weight of wet soil
- (iii) Weight of dry air
- (iv) Volume of air voids

(b) If the moisture content of a soil is 20% and the dry weight is 100g, what is the weight of water?
- (i) 10g
- (ii) 20g
- (iii) 25g
- (iv) 30g

(c) Plasticity of soil refers to the
- (i) Soil's ability to retain moisture
- (ii) Soil's ability to undergo deformation without cracking
- (iii) Soil's resistance to deformation
- (iv) Soil's permeability

(d) A soil sample has a shrinkage limit of 10% and specific gravity of soil solids of 2.7. The porosity of the soil at shrinkage limit is
- (i) 21.2%
- (ii) 52.7%
- (iii) 30%
- (iv) 70%

(e) Which instrument is commonly used to determine liquid limit?
- (i) Proctor apparatus
- (ii) CBR apparatus
- (iii) Casagrande apparatus
- (iv) Core cutter

(f) Darcy's law states that the velocity of flow is directly proportional to
- (i) Area
- (ii) Hydraulic gradient
- (iii) Time
- (iv) Soil weight

(g) Maximum dry density is achieved at
- (i) Zero moisture
- (ii) Optimum moisture
- (iii) Air-dried condition
- (iv) High moisture

(h) Contact pressure under a rigid footing is generally
- (i) Parabolic
- (ii) Uneven
- (iii) Sinusoidal
- (iv) Uniform

(i) The ratio of volume of water present in a given soil mass to the total volume of its void is known as
- (i) Porosity
- (ii) Percentage voids
- (iii) Void ratio
- (iv) Degree of saturation

(j) Permeability is highest in
- (i) Clay
- (ii) Silt
- (iii) Silty clay
- (iv) Gravel

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Q.2 Solve both questions : **[7+7=14]**

(a) What is meant by a three-phase soil system? Draw a neat phase diagram showing weight and volume relationships.

(b) A saturated clayey soil sample was tested in the laboratory. It has a specific gravity (G) of 2.70 and a water content (w) of 25%. Determine the: (i) Void ratio (e) (ii) Degree of saturation (S)

Q.3 Solve both questions : **[7+7=14]**

(a) Explain the significance of the liquid limit, plastic limit, and shrinkage limit in soil mechanics. Sketch the Atterberg limits diagram.

(b) Two soil samples A and B have the following properties: (i) Soil A: LL=60%, PL=40%, Clay =35% (ii) Soil B: LL=60%, PL=30%, Clay =50%. Compare their plasticity indices and activities, and identify which is more problematic in terms of swelling/shrinkage.

Q.4 Solve both questions : **[7+7=14]**

(a) Compare the constant head and falling head permeability tests in terms of soil suitability and test setup.

(b) Water flows through a soil sample of length 20 cm and cross-sectional area 50 cm^2. The head loss measured is 10 cm over 5 minutes and the discharge is 300 cm^3. Calculate the coefficient of permeability.

Q.5 Solve both questions : **[7+7=14]**

(a) State Darcy's Law. What are its assumptions and limitations? Under what soil conditions is Darcy's law not valid?

(b) A soil profile consists of a 6 m thick homogeneous soil layer with the water table located at a depth of 3 m from the ground surface. The unit weight of soil above the water table is 18 kN/m^3 and the saturated unit weight below the water table is 20.5 kN/m^3. Assume γ_w= 9.81 kN/m^3. Determine the effective stress at a depth of 6 m below the ground surface. Illustrate your solution with a stress distribution diagram, showing total stress, pore water pressure, and effective stress with depth.

Q.6 Solve both questions : **[7+7=14]**

(a) Differentiate between the Standard Proctor Test and the Modified Proctor Test for soil compaction. Explain their objectives and test procedures.

(b) In a Standard Proctor test, a maximum dry density of 1.85 g/cm^3 is obtained at 12% water content. What is the dry density at 14% moisture content if bulk density is 2.00 g/cm^3?

Q.7 Solve both questions : **[7+7=14]**

(a) State Boussinesq's assumptions for stress distribution in a semi-infinite soil mass. When is this theory not valid?

(b) Differentiate between contact pressure under rigid and flexible footings. How does the distribution vary?

Q.8 Solve both questions : **[7+7=14]**

(a) State the assumptions of Terzaghi's 1D consolidation theory. Discuss its limitations in real field conditions.

(b) Derive the relationship between: (i) Void-ratio and porosity (ii) Moisture content and degree of saturation

Q.9 Solve both questions : **[7+7=14]**

(a) Differentiate between compaction, normally consolidated and over consolidated soils. How would you determine the over consolidation pressure?

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(b) A concrete dam is constructed across a river over a permeable stratum of soil of limited thickness. The water heads are upstream side 16 m and 2 m on the downstream side. The flow net constructed under the dam gives N_f=7 and N_d=12. Calculate the seepage loss through the subsoil if the average value of the hydraulic conductivity is 6×10^{-3}cm/sec horizontally and 3×10^{-4}cm/sec vertically. Calculate the exit gradient if the average length of the last field is 0.9 m.

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